

Kanav Sabharwal

Email: kanav.sabharwal@u.nus.edu

Phone: +65-9673-4707

Web: kanavsabharwal.github.io | [Google Scholar](#)

LinkedIn: linkedin.com/in/kanav-sabharwal

RESEARCH INTERESTS

Advancing wireless networking systems using AI techniques to enhance Physical Layer capabilities, improve performance, and ensure security.

EDUCATION

- National University of Singapore** Jan 2022 - Present
 - Doctor of Philosophy (Ph.D.) - Computer Science
 - GPA: 5/5
 - Advisor: [Dr. Mun Choon Chan](#) and [Dr. Dinil Mon Divakaran](#)
 - Research Direction: Focussing on AI-enhanced physical layer for wireless networks
 - Graduate Courses: Advanced Topics in AI, Network Security, Advanced Topics in Networking, Exploration in CS Research
- National University of Singapore** Aug 2020 - Dec 2021
 - Master of Computing (AI Specialization)
 - GPA: 4.83/5
 - Advisor: [Dr. Dinil Mon Divakaran](#)
 - Thesis: Bandwidth Tunable Defence to Improve IoT network Security using Adversarial Machine Learning
 - Graduate Courses: AI Planning and Decision Making, Uncertainty Modeling in AI, Neural Networks and Deep Learning, Natural Language Processing, Information Visualisation, Knowledge Discovery and Data Mining
- Vellore Institute of Technology, Vellore** Jul 2016 - Jul 2020
 - Bachelor of Technology (Information Technology)
 - GPA: 9.05/10

CONFERENCE PUBLICATIONS

- Enhancing LoRa Reception with Generative Models: Channel-Aware Denoising of LoRaPHY Signals**
[SENSYS '24] [\[Paper\]](#) [\[Code\]](#)
Kanav Sabharwal, Soundarya Ramesh, Jingxian Wang, Dinil Mon Divakaran, Mun Choon Chan
- EGAL: Enhancing LoRa Network Lifetime with Load Balancing**
[SECON '24]
Malaika Afra Taj, Kanav Sabharwal, Mun Choon Chan
- Attacking logo-based phishing website detectors with adversarial perturbations**
[ESORICS '23] [\[Paper\]](#)
Jehyun Lee, Zhe Xin, Melanie Ng Pei See, Kanav Sabharwal, Giovanni Apruzzese, Dinil Mon Divakaran
- Testing Masks and Air Filters With Your Smartphones**
[SENSYS '23] [\[Paper\]](#)
Bangjie Sun, Kanav Sabharwal, Gyuyeon Kim, Mun Choon Chan, Jun Han
- iPET: privacy enhancing traffic perturbations for secure IoT communications**
[PETS '23] [\[Paper\]](#) [\[Code\]](#)
Akshaye Shenoi, Prasanna Karthik Vairam*, Kanav Sabharwal*, Jialin Li, Dinil Mon Divakaran*
**Authors contributed equally*

SKILLS

- Technical Skills:** Machine Learning (Supervised, Unsupervised, and Deep Learning), Generative AI, Signal Processing, Wireless Networks, Physical Layer Design, Software Defined Radio (SDR)
- Languages and Frameworks:** Python, C/C++, PyTorch, TensorFlow, GNURadio, MATLAB, NumPy, SciPy
- Soft Skills:** Critical Thinking, Problem Solving, Research, Team Collaboration, Leadership, Technical Writing, Fast Learner and Adaptability

EXPERIENCE

- **Student Researcher** Nov 2020 - Jul 2021
NUS-Singtel Cyber Security R&D Lab, Singapore
 - Researched and developed an innovative defence mechanism to enhance IoT network security through Adversarial Machine Learning techniques
 - Implemented advanced device fingerprinting attack models to assess defence's effectiveness across various scenarios. The defence mechanism with a tunable bandwidth overhead, substantially improves privacy compared with existing approaches
 - Published research findings as primary author to an A-rank conference
- **Machine Learning Intern** Dec 2019 - Jun 2020
6D Technologies, Bangalore, India
 - Collaborated with the R&D team to develop an ML-based 'SMS Firewall' capable of classifying spam and A2P messages within the network with 98% accuracy, while minimizing false positives—a critical aspect given the nature of the application
 - Developed a 'SIM Box Detector' by optimizing the analysis of network statistics
- **Academic Intern** Jun 2019
Hewlett Packard Enterprise (HPE) and NUS, Singapore
 - Ranked among the top students in the academic program focused on 'Big Data Analytics using Artificial Neural Networks.' Led a team of 5 members on various group projects as part of the curriculum. Received a Letter of Recommendation (LOR) for outstanding performance
- **Summer Intern** May 2018 - Jun 2018
Bharat Electronics Limited, Ghaziabad, India
 - Proposed a Machine Learning based substitute to the present Radar Surveillance system while studying about the role of IT in defense sector

TEACHING EXPERIENCE

- Teaching Assistant : **CS5229**- Advanced Computer Networks w/ [Dr. Mun Choon Chan](#)
Fall 2022, Fall 2023, Fall 2024
- Teaching Assistant : **CS5422/4222**- Wireless Networking w/ [Dr. Ambuj Varshney](#)
Spring 2023, Spring 2024
- Teaching Assistant : **CS5346**- Information Visualisation w/ [Dr. Bimlesh Wadhwa](#)
Spring 2022

PROFESSIONAL SERVICE

- **Contributed Reviews (Sub-Reviewer)**
Conferences: ICNP 2023, SENSYS 2024
- **Artifact Evaluation Committee**
Conferences: SENSYS 2024

HONORS AND CERTIFICATIONS

- Graduate Research Scholarship, National University of Singapore
- Letter of Recommendation from the Assistant Vice President of Product Development at 6D Technologies, recognizing contributions to the R&D team in developing an AI-based 'SMS Firewall'.
- GAIP Scholarship for ranking among the top students in the Global Academic Internship Program 2019 at the National University of Singapore. Received a Letter of Recommendation from HPE for outstanding performance and a Letter of Evaluation from NUS with an A+ grade.
- Coursera Courses Completed: Machine Learning by Andrew Ng | Neural Networks and Deep Learning | Text Retrieval and Search Engines | Cloud Computing Applications, Part 1: Cloud Systems and Infrastructure
- Merit-based scholarship across all 8 semesters at VIT Vellore.
- Gold Medal awarded for being top performer (scholar) throughout senior secondary school.

EXTRA CURRICULAR ACTIVITIES

- Core Member, Indian Society for Technical Education: Managed technical events, led a team of volunteers, and secured funding for various initiatives.
- Core Member, FEPSI (NGO): Mentored underprivileged students by combining academic programs with engaging activities as part of outreach efforts.
- Coordinated technical events and fests, including Horizon'17 and GraVITas'17. Volunteered for the Events and Cultural Committee at Riviera'18.